

Activity API — What Changed

For mobile / app developers integrating the Activity Score & Analytics endpoint

Endpoint: GET /api/wearable/activity/analytics | **Branch:** Staging | **Files touched:** activityAnalyticsService.js, dailyHealthScoreService.js, exerciseGuidanceService.js, activityScoreInsightService.js (new)

This document explains, in plain terms: what problem existed before, what was changed on the backend, and exactly where the new fields should be used on the app side. Read this before wiring up the Activity screen.

1. The Problem We Fixed

A user's daily activity can come from two places: a connected wearable device (Apple Watch, Fitbit, etc. via background sync) and a manually logged workout (POST /api/exercise/log). Both sources can describe the **same real-world workout** — e.g. the watch detects a 30-minute run via elevated heart rate AND the user separately logs the same run.

Before this change, the Activity Score counted BOTH signals separately:

- Device's activeMinutes → counted toward the score
- Manual's cardioMinutes → also counted toward the score, as a separate signal

Result: the same 30-minute workout could inflate the score twice (once as "device movement", once as "manual movement") — a real double-counting bug.

2. Before vs After

2.1 Scoring logic

Signal	BEFORE	AFTER
Steps	Device only	Device only — unchanged
Movement / Cardio minutes	Device activeMinutes AND manual cardioMinutes scored as two separate signals → could double-count the same workout.	Merged into ONE signal: movementMinutes. Device wins if it reported anything that day; manual is used ONLY as a fallback when the device reported nothing. Never both.
Strength sessions	Manual only, rolling 7-day count	Unchanged — manual only, rolling 7-day count
Number of scoring signals	Up to 4 (steps, device-movement, manual-movement, strength)	Exactly 3 (steps, movementMinutes, strength) — this is WHY the same day's score changed after the fix (e.g. 75 → 66.7), it is the correct number now.
Calories (weekly/monthly average)	device.caloriesBurned + manual.caloriesBurned — always added .	Device-priority / manual-fallback, same rule as movement — never added , so a synced workout's calories aren't counted twice.

2.2 Response shape

Two fields were added to every daily entry, and are also present in the weekly/monthly summary[] buckets:

- movementMinutes: {{ actual, goal, source }} — the single number that fed the score's movement signal, and which source ("device" or "manual") it came from.
- insight: {{ band, headline, overallSummary, topGaps, recommendations, disclaimer }} — a rule-based (non-AI) explanation of the score, same pattern as the existing Nutrition Insight.

3. New Response Shape — Daily

GET /api/wearable/activity/analytics?range=daily&date=2026-09-25

```
{
  "success": true,
  "range": "daily",
  "startDate": "2026-09-25",
  "endDate": "2026-09-25",
  "hasWearableConnected": true,
  "goals": {
    "steps": 7500,
    "activeMinutes": 21.4,
    "cardioMinutes": 21.4,
    "strengthSessionsPerWeek": 2
  },
  "days": [
    {
      "date": "2026-09-25",
      "device": {
        "steps": 8000,
        "activeMinutes": 52,
        "caloriesBurned": 650
      },
      "manual": {
        "cardioMinutes": 0,
        "totalExerciseMinutes": 45,
        "sessionsCount": 1,
        "caloriesBurned": 280,
        "strengthSessionsLast7Days": 1
      },
      "movementMinutes": { "actual": 52, "goal": 21.4, "source": "device" },
      "goals": { "steps": 7500, "activeMinutes": 21.4, "cardioMinutes": 21.4, "strengthSessionsPerWeek": 2 },
      "activityScore": 83.3,
      "calendarBand": "good",
      "insight": {
        "band": "good",
        "headline": "Good activity day",
        "overallSummary": "Good activity day - your Activity Score is 83/100. Short on Strength.",
        "topGaps": [
          { "signal": "strengthSessions", "label": "Strength", "source": "manual",
            "message": "Strength was 50% of your target (1 vs 2)." }
        ],
        "recommendations": [
          { "action": "Get in a logged strength/resistance session - gym, bodyweight training, or resistance bands",
            "estimatedPointGain": 16.7 }
        ],
        "disclaimer": "This is a wellness insight based on your logged activity, not medical or fitness-coaching advice."
      }
    }
  ]
}
```

Important: insight is NOT always present

insight is only computed for a true single-day lookup (?date= or a one-day range). For a multi-day range (e.g. a full calendar month), each day in days[] will have activityScore and calendarBand only — insight will be undefined. This is deliberate (performance — the calendar view only needs the score dot, not 30 rule-based computations per request). **Do not assume insight exists — always check for it before rendering.**

4. Response Shape — Weekly / Monthly / Yearly

Same call with range=weekly / monthly / yearly returns summary[] instead of days[]. Each bucket's insight is built from that period's AVERAGES, not any single day — read it as "on an average day this period".

```
{
  "success": true,
  "range": "weekly",
  "startDate": "2026-09-21",
  "endDate": "2026-09-26",
  "hasWearableConnected": true,
  "goals": { "steps": 7500, "activeMinutes": 21.4, "cardioMinutes": 21.4, "strengthSessionsPerWeek"
: 2 },
  "summary": [
    {
      "period": "2026-W39",
      "daysLogged": 5,
      "avgSteps": 7200,
      "avgActiveMinutes": 18,
      "avgCardioMinutes": 30,
      "avgCaloriesBurned": 950,
      "avgActivityScore": 62,
      "insight": { "band": "low", "headline": "...", "overallSummary": "...", "topGaps": [ ], "reco
mmendations": [ ] }
    }
  ]
}
```

5. Field-by-Field Cheat Sheet — What To Use Where

Field	Where to use it on the app	Notes
<code>activityScore</code>	Ring/gauge on the Activity screen, calendar dot value	Single source of truth — same number the Health Score uses internally. Never recompute on device.
<code>calendarBand</code>	Color of the calendar dot: optimal/good/low	optimal ≥ 85 , good ≥ 70 , low below 70
<code>movementMinutes.actual/.goal</code>	Progress bar for "Movement" / "Cardio" tile	This IS the number that fed the score — do not also show <code>device.activeMinutes</code> and <code>manual.cardioMinutes</code> as if they were two separate progress bars, that recreates the double-count visually.
<code>movementMinutes.source</code>	Small badge/icon: "from your watch" vs "from your logged workout"	Purely informational, does not affect the number shown.
<code>device.steps</code>	Steps tile	Always device-only. If <code>hasWearableConnected</code> is false, treat as "no data", not "0 steps".
<code>manual.strengthSessionsLast7Days</code>	Strength tile actual value	Rolling 7-day count, not "did they train today" — a rest day is expected and correct.
<code>goals.*</code>	Target values / max of progress bars	Personalized by age + goal type. Same for every day in the response — fetch once, do not expect it to change per day.
<code>insight.overallSummary</code>	Headline text at the top of the Activity detail screen	Human-readable one-liner, safe to render directly.
<code>insight.topGaps[]</code>	"Where you fell short" list/cards	Up to 3 items, worst signal first. Each has a ready-to-render message.
<code>insight.recommendations[]</code>	"What to do next" cards, sorted best-impact first	<code>estimatedPointGain</code> is a real computed number — safe to show as "+X points".
<code>insight.disclaimer</code>	Small print under any insight card	Must be shown wherever insight text is shown — not medical/coaching advice.
<code>hasWearableConnected</code>	Whether to show a "connect a device" prompt	Top-level field, applies to the whole response, not per-day.

6. Do / Don't for the App Team

Do

- Render `movementMinutes` as ONE progress bar, using its own source for the badge.
- Always null-check `insight` before rendering (see Section 3).
- Treat `activityScore` as read-only, coming from the backend.
- Cache-bust: this endpoint is server-side cached for 5 minutes per user/range/date — if you log a new workout and immediately re-fetch, the score may take up to 5 minutes to reflect it.

Don't

- Don't add `device.activeMinutes` and `manual.cardioMinutes` together in the UI — that recreates the exact bug this change fixed.
- Don't add `device.caloriesBurned` and `manual.caloriesBurned` together for a period total — use the backend's `avgCaloriesBurned` instead, which already applies the same device-priority rule.
- Don't assume `insight` is always present.

7. Known Limitation (Not Yet Fixed)

If a workout is synced from a wearable (e.g. Apple Watch) via `POST /api/exercise/log` (rather than the OS-level background sync), it is currently still labeled `source: "manual"` in the response, because the backend has no explicit signal telling it the log came from a device. The NUMBERS are correct either way (no double-counting) — only the source label can be misleading in this one case. A fix is planned (either the app sending an explicit `origin` field, or routing device workouts through the OS-sync endpoint instead). Do not build UI logic that depends on source being 100% accurate for Apple Watch strength/cardio sessions until this is resolved.